

Bangyen Pham

Backend Developer, AI/ML Engineer

Chicago, IL

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<https://bangyen.github.io>

EDUCATION

Northwestern University, MS — *Computer Science*

SEPTEMBER 2022 - MARCH 2024

University of Virginia, BA — *Mathematics*

AUGUST 2018 - MAY 2021

SKILLS

- **Programming & Tools:** Python, C/C++, JavaScript, Docker, Terraform
- **ML/AI:** PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas
- **Systems & Cloud:** AWS/GCP, Linux, HPC
- **Databases & Backend:** SQL/NoSQL, FastAPI, Flask, Node.js

EXPERIENCE

Volta Health — *Product Engineer*

JULY 2024 - AUGUST 2025

- Built an AI-powered search platform for university resources using OpenAI technology, achieving 94% search accuracy and processing hundreds of student queries with automated content filtering
- Designed a scalable cloud architecture with auto-scaling capabilities and infrastructure automation, reducing deployment time by 40% and achieving high system reliability
- Created a location-based healthcare facility finder integrating mapping services, helping students discover 200+ nearby providers across 7 healthcare categories with 60% faster resource discovery

Center for Nuclear Femtography — *HPC Engineering Intern*

MAY 2023 - SEPTEMBER 2023

- Designed and implemented a C++ translation of a Fortran-based computational method, using parallel processing to reduce computation time by 5% on a dataset of 10,000 simulations
- Developed and optimized code integration and debugging workflows within simulation frameworks, ensuring 99.5% accuracy in nuclear femtography simulations

PROJECTS

ZSharp — *Sharpness-Aware Minimization*

Reproduced the ZSharp algorithm in PyTorch with Apple Silicon optimization, delivering +5.2% accuracy over SGD, a 4.4× training speedup, and complete reproducibility with full test coverage

Oligopoly — *Agent-Based Economic Modeling*

Built a comprehensive FastAPI and SQLAlchemy-based simulation platform supporting 2-4 firms across 20+ rounds with collusion detection, policy shock modeling, and real-time heatmap analysis

PUBLICATIONS

Generalized Collective Algorithms for the Exascale Era — *CLUSTER 2023*

Revisiting Computation for Research: Practices and Trends — *SC 2024*